Valorization of Clitoria ternatea Biomass for Domestic Wastewater Treatment and Sustainable Concrete Production: A Circular Water-Energy Nexus Approach

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01 INTRODUCTION

Freshwater scarcity and domestic wastewater generation necessitate sustainable water reuse strategies. Biochar-based treatment offers an environmentally friendly approach for wastewater purification and resource recovery.

This study utilizes **Clitoria ternatea-derived biochar (CT-BC)** to treat domestic wastewater and investigates the reuse of the treated water in M30 concrete. The mechanical and microstructural performance of concrete were evaluated to assess its feasibility for sustainable construction.

02 OBJECTIVES

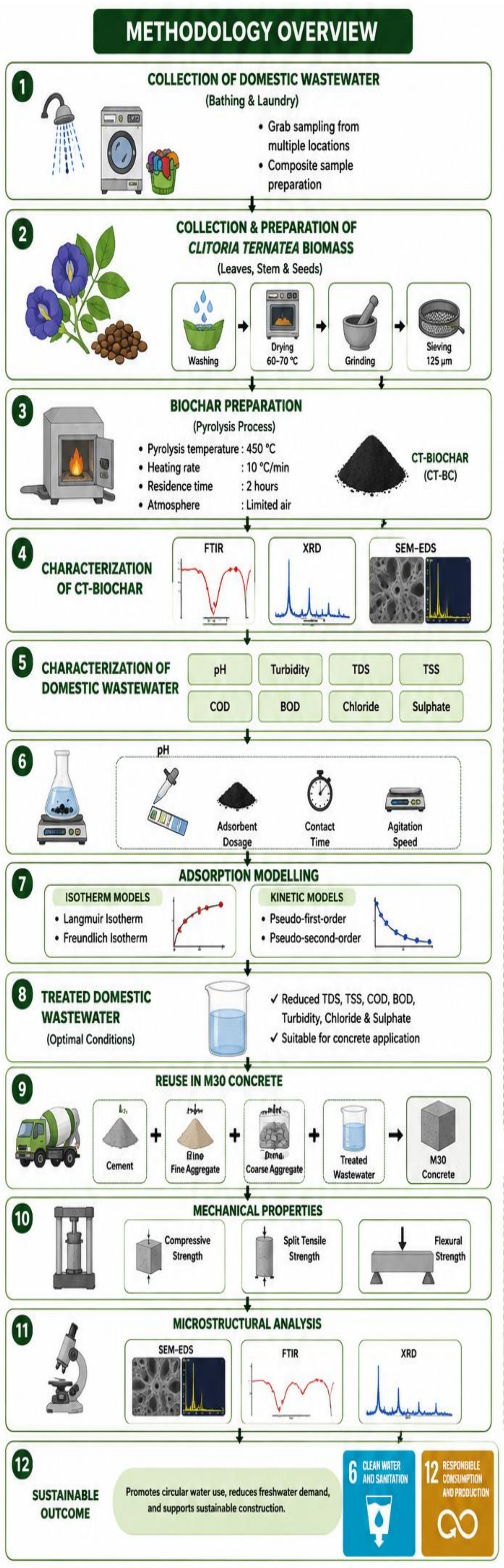
✓ Synthesize and characterize **Clitoria ternatea-derived biochar (CT-BC)**.

✓ Treat domestic wastewater and evaluate adsorption performance.

✓ Reuse treated wastewater in **M30 concrete**.

✓ Assess the mechanical and microstructural properties of concrete.

03 METHODOLOGY



04 RESULTS

The treatment efficiency of CT-BC and the mechanical performance of concrete prepared with treated domestic wastewater are presented.

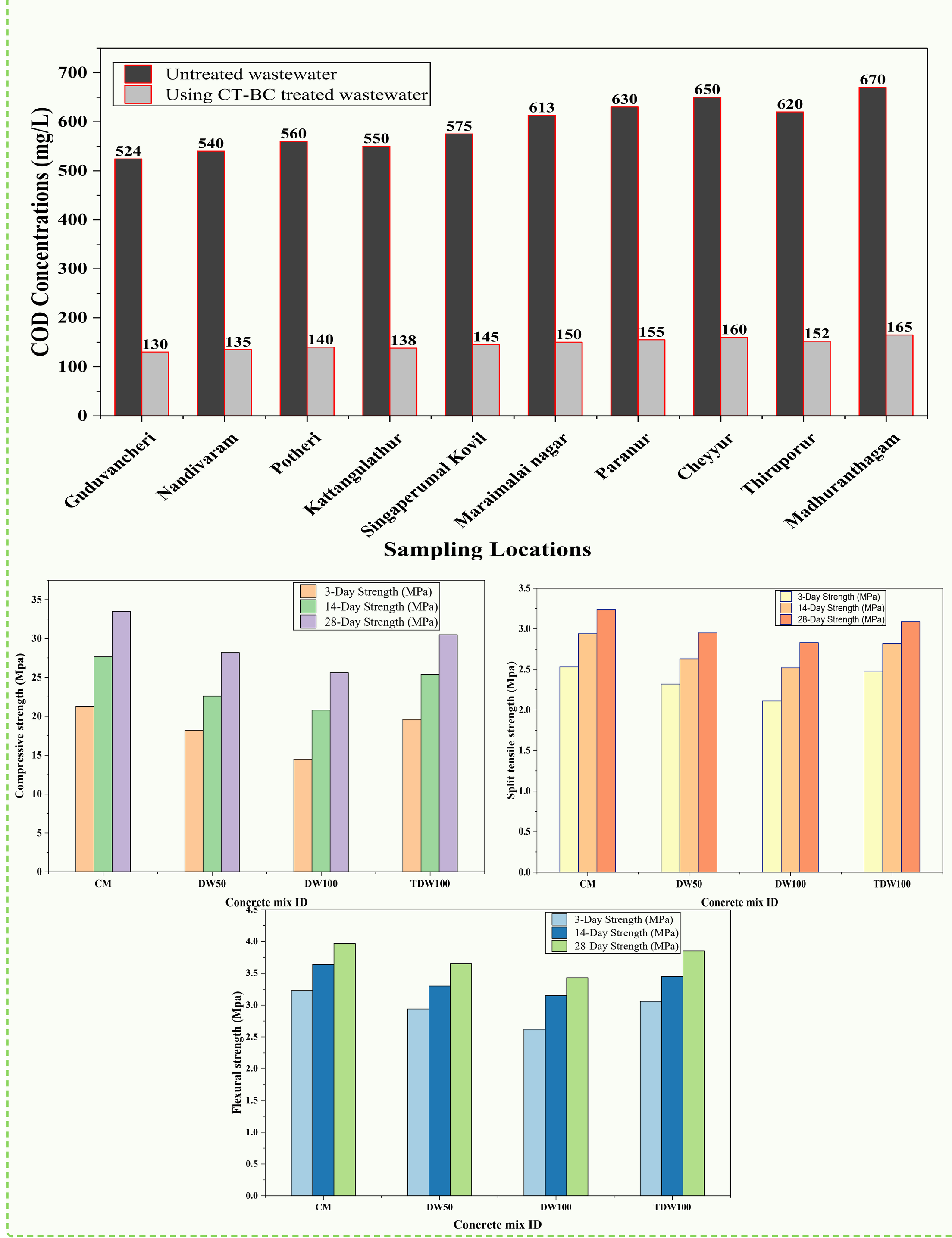
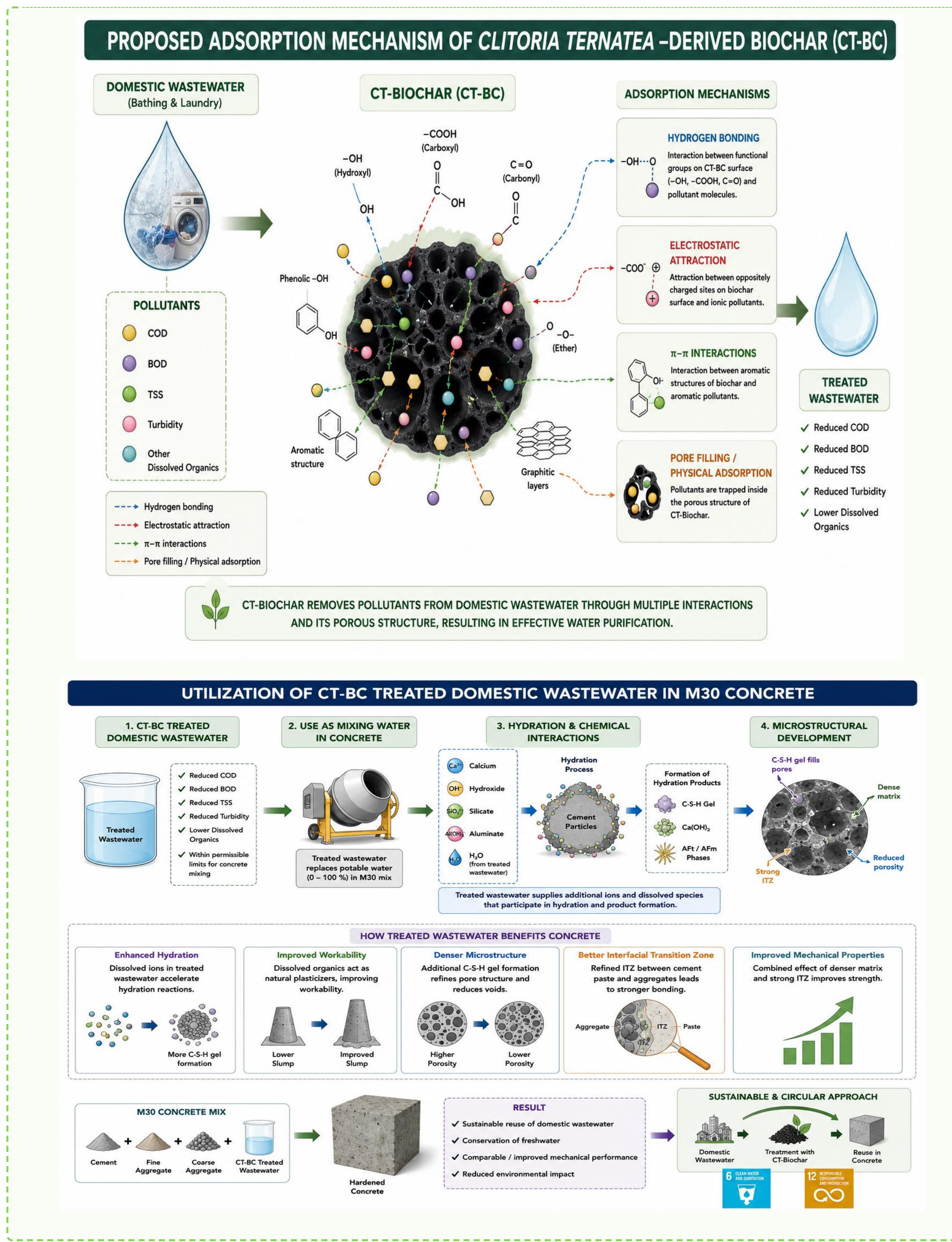


Figure 2. Proposed mechanism for CT-BC-treated wastewater reuse in concrete.

S.no	Physical Properties	Range	Analytical Procedures
1.	Colour	Black	Present study
2.	pH	6.9	IS: 3025 (Part 11) – 2022
3.	Moisture content (%)	6.1	IS: 1350 (Part 1) – 1984
4.	Bulk Density (g/cm^3)	0.40	ASTM D2854
5.	Specific gravity	1.32	IS: 2720 (Part 3) – 1980
6.	Adsorption capacity	58 mg/g	Present study

Table 1. Physical parameters of CT biochar



Scheme 1. Proposed reaction pathway.

05 DISCUSSION

The combined adsorption and reuse approach demonstrated that CT-BC-treated domestic wastewater can effectively replace potable water in concrete without compromising performance. This integrated strategy supports circular water management and sustainable construction practices.

Figure 3. Sustainable reuse of CT-BC-treated domestic wastewater in concrete production.

06 CONCLUSIONS

- CT-BC effectively reduced key pollutants from domestic wastewater through adsorption, demonstrating its potential as a sustainable bio-based adsorbent.
- Concrete prepared with CT-BC-treated domestic wastewater exhibited satisfactory mechanical and microstructural performance, comparable to conventional concrete.
- The integrated wastewater treatment and reuse approach promotes freshwater conservation and supports circular economy principles in sustainable construction.

"Wastewater is not a waste—it is a valuable resource when treated with CT-BC for sustainable concrete production."

07 FUTURE WORK

Evaluate the long-term durability and life-cycle performance of concrete produced with CT-BC-treated domestic wastewater. Develop continuous-flow treatment systems and investigate the large-scale application of CT-BC for sustainable wastewater reuse in construction.

08 REFERENCES

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09 ACKNOWLEDGEMENTS

The authors gratefully acknowledge the Department of Civil Engineering, SRM Institute of Science and Technology, Kattankulathur, for providing laboratory facilities and research support. The authors also thank the faculty members and laboratory staff for their valuable guidance and technical assistance throughout this study.